FLOODGAUGE · RISK REGISTER

FloodGauge risk register and hazard analysis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Initial draft for the kit 1.9.1 upgrade; project risks, preliminary hazard analysis and first-pass design FMEA of option A with notes for B and C

FloodGauge risk register and hazard analysis

Top risks: the radar module is unverified, the drain head seal is only IP67, and the pilot city has not been approached. The main hazards are an arm falling onto a road and work at height beside traffic; both are controlled by the through-bolt and a traffic plan. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection is scored 1 (certain to be caught) to 10 (not caught) in the FMEA. The scores are first-pass judgements by the design lead for review with a partner; they are not measured.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
R-1	Radar module data sheet does not support the 2 mm accuracy or the lens beam	Accuracy and beam width are assumed (CAL Table 1)	Street depth error above 10 mm (PR-002) or a false curb echo	3	4	Design lead	Read the data sheet before buying; if it fails, use option B's pressure transducer
R-2	Pilot city refuses an arm over the gutter or an attachment on the pole	No road authority has been approached	Option A cannot be installed; stand pipe options B or C needed	3	5	Pilot partner	Ask early; FLG-OPT-001 carries B as the fallback
R-3	Drain head loses its seal in a long flood	Transducer face keeps the stock IP67 seal (PR-009)	Basin reading lost during the event that matters most	3	4	Design lead	Ask the maker for a rating; consider option B's sealed stainless transducer for the basin
R-4	Mass on the pole exceeds 5 kg	Margin is 0.01 kg on model volumes (PR-015)	Pole owner rejects the load	3	3	Design lead	Weigh at TRL 4; 2.5 mm bracket plate is the named fix
R-5	Airtime exceeds network rules at slow spreading factors	SF10 and slower (PR-007)	Network operator limits or blocks the gateway	3	3	Network operator	Private or city gateway for event reporting (FLG-DEC-001)
R-6	Cost stays above the USD 160 target	Radar and arm are USD 90 of USD 205.50	Fewer gauges per budget	4	2	Design lead	Quote the radar; value-engineering list in DEC
R-7	Tampering or theft at the sidewalk	Fixings within reach (PR-018)	Gauge removed or damaged	3	3	Pole owner	Security fixings are in the design; option A keeps the arm out of reach
R-8	A fixed 60 GHz radar is not covered by the module's radio certificate	G1 and G2 are to assess	Cannot be used legally in the pilot country	2	5	Design lead	Check the certificate scope before the pilot
R-9	Pilot partner never agrees to host a gauge	No partner is agreed (PRB open questions)	Design stays on paper	3	4	Project lead	Approach a county flood control district (FLG-DEC-001)

Preliminary hazard analysis

Table 2. Hazards. Severity is 1 (minor) to 5 (fatal); likelihood is 1 (remote) to 5 (frequent).

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H-1	Arm or head falls onto the roadway	Clamp slip, bolt failure or vandalism	Strike to a person or vehicle	5	2	Through-bolt and two clamps, factor 37 against wind twist; 50 kg hang case passes (CAL H)	Low if the through-bolt is fitted
H-2	Head or arm struck by a tall vehicle	Lens 4.6 m above a 4.3 m vehicle	Damage to the vehicle or the gauge	3	2	Lens set per site with 0.3 m margin (PR-001)	Low
H-3	Person trips on the surface cable covers	Covers on the gutter and sidewalk are 16 mm high	Fall	3	3	Low hat section with ramped ends; edges broken (PR-019); marker plate	Medium; to review with the partner
H-4	Cut from a sharp edge during installation or by a passer-by	Cut sheet metal and bolt ends	Laceration	2	3	Edges broken in the model; covers 2 mm folded; guards on bolt ends to assess	Low
H-5	Fall or strike while lifting the arm to 4.6 m	Work at height beside traffic	Serious injury	5	2	Two-person crew with traffic protection, mobile platform (CAL L1); not tested at TRL 3	Medium; training and traffic plan needed
H-6	Gas, drowning or crush when lifting a grate or entering a basin	Confined space and heavy grate	Serious injury	5	2	No entry; grate lifted with a hook by two people (PR-012)	Low
H-7	Lithium iron phosphate cell overheats in sun	Hot enclosure above the charge limit (FieldNode)	Fire or cell damage	4	2	Charger cut-out and the cell chemistry; FieldNode safety case	Medium; inherited from FieldNode
H-8	Radio emission exposure	60 GHz and 868 MHz low power	None expected at the power levels of certified modules	1	1	Certified modules; to assess (REG G1, G2)	Low

Design FMEA (first pass, P3)

Table 3. Failure modes of option A (rows F-1 to F-12) and of the new parts in options B and C (F-13, F-14). RPN = severity x occurrence x detection, each scored 1 to 10 here; a row with an RPN above 100 gets an action before TRL 4.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F-1	Radar head	Measure street range	No echo or false echo from the curb	Wrong or missing street depth	Rain on the PTFE lens; curb inside the beam	4	4	5	80	Record a dry-road background; reject the curb echo (CAL G2)
F-2	Radar head housing	Keep water out	Seal leak at the lens or gland	Radar fails	Gasket creep, UV	7	3	4	84	Specify a sealed lid; check the housing rating
F-3	Drain head ranger	Measure basin level	Echo from the tube wall or condensation	Reading error	69 mm bore; fog on the face	5	4	5	100	Use first-echo logic and the NTC; test at TRL 4
F-4	Drain head seal	Survive submersion	Water reaches the transducer	Drain reading lost	IP67 face only (PR-009)	7	4	6	168	Ask for a rating; consider option B transducer
F-5	Stilling tube slots	Pass water, damp waves	Slots clog with leaf litter	Level lags the basin	Biofilm and debris	5	4	4	80	Clean at each grate visit; the open area has margin (STM S21)
F-6	Pole bracket and band clamps	Hold the arm	Slip or twist on the pole	Head moves, datum lost	Preload loss, vibration	7	3	4	84	Specify clamp torque for 1,000 N (DEC to confirm 4)
F-7	M8 through-bolt	Stop twist	Shear or bearing failure in a thin pole wall	Arm swings round	2.5 mm wall assumed (DEC item 5)	8	2	5	80	Confirm pole wall at each site (DEC to confirm 5)
F-8	Arm tube	Carry the head	Fatigue crack at the cleats	Head falls	Wind buffeting, 1.6 mm wall	9	2	5	90	Fillet-free cleats; inspect yearly
F-9	Drain cable	Carry data	Cable cut or crushed under the cover or by the grate	Drain reading lost	Grate seated on the cable	6	4	4	96	Slack loop tied to the tube clamp; route by the grate opening
F-10	Surface covers	Protect the cable	Cover lifted or tripped on	Cable exposed; trip hazard	Anchor removed	6	3	3	54	Security nuts on anchors
F-11	FieldNode (shared)	Power and radio	Cell does not charge in heat or cold	No data	Charge window 0 to 45 C	6	4	5	120	Inherited; see FieldNode risks
F-12			Bands fade		UV on film	3	5	3	45	

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
	Marker plate	Show depth by eye		Misread depth						Use UV-stable film; yearly check
F-13	Option B transducer	Measure pressure	Zero shifts with a wet vent	Constant depth error	Condensation in the vent tube (STM B3)	6	4	6	144	Desiccant or membrane; yearly zero check
F-14	Option C float	Move with the water	Float binds or fills	Reading sticks	Silt, ice or a cracked shell (STM C5)	6	4	5	120	Hollow shell fix made (FLG-DDR-004 F1); stop collars

Safety

Work at height beside traffic, lifting a grate, a pole-mounted mass, a lithium iron phosphate cell and mains-free 12 V electronics are the hazards in this design. No mains voltage is present. The installation sequence is paper work only at TRL 3; no one has built or installed a unit.